

as the subspace spanned (or generated) by $v_1, \dots, v_p$. $\square$

Zero subspace: Set consisting of only the zero vector in $\mathbb{R}^n$.

Column Space and Null Space of a Matrix

Definition: The column space of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .

If $A = [a_1 \dots a_n]$, with the columns in $\mathbb{R}^m$, then $\text{Col } A$ is the same as $\text{Span}\{a_1, \dots, a_n\}$.

Example 4: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$. Determine whether b is in the column space of A .

Solution: By Defn, we need to verify whether b is a linear combination of the columns of A , i.e. whether $\exists c_1, c_2, c_3 \in \mathbb{R}$ s.t. $c_1 a_1 + c_2 a_2 + c_3 a_3 = b$ where a_1, a_2 and a_3 are the 1st, 2nd and 3rd columns of A respectively.

By Defn. of Ax , we have $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = b$.

By Theorem 3 from Chapter 1, $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = b$ has the same solution set as the system of equations corresponding to the augmented matrix

$$\begin{aligned} & \begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 4 \times \text{Row 1}}]{\substack{\text{Row 3} = \text{Row 3} \\ - 1 \times \text{Row 1}}} \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ -3 & 7 & 6 & -4 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-1/6)}]{\substack{\text{Row 3} = \text{Row 3} \\ + 3 \times \text{Row 1}}} \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & 1 & 3 & 5/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

It's very easy to verify that the final matrix is in echelon form. By Theorem 2 from Chapter 1, there are no rows in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. $\therefore$ The system of equations